

Purification Of Organic Compounds

Questions with Answer Keys

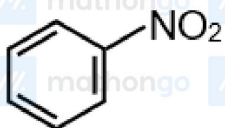
JEE Main 2020 Chapterwise

Chemistry

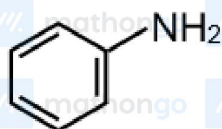
Q1 JEE Main 2020 - 8 January (Evening)

Kjeldahl's method cannot be used to estimate nitrogen for which of the following compounds?

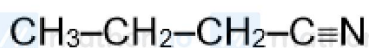
(A)



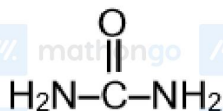
(B)



(C)

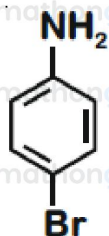


(D)



Q2 JEE Main 2020 - 2 September (Morning)

In Carius method of estimation of halogen, 0.172g of an organic compound showed presence of 0.08g of bromine. Which of these is the correct structure of the compound?



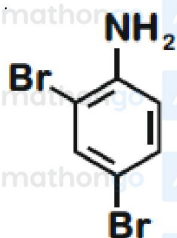
(A)

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Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry



- (B)
(C) $\text{H}_3\text{C} - \text{Br}$
(D) $\text{H}_3\text{C} - \text{CH}_2 - \text{Br}$

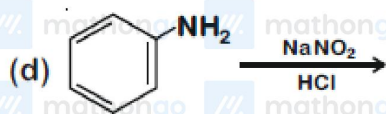
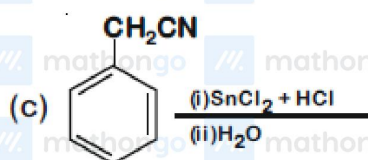
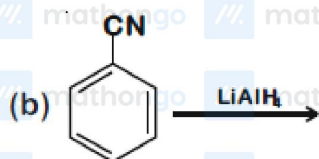
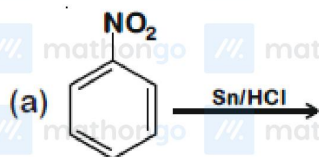
Q3 JEE Main 2020 - 3 September (Morning)

Glycerol is separated in soap industries by

- (A) Fractional distillation
(B) Differential extraction
(C) Distillation under reduced pressure
(D) Steam distillation

Q4 JEE Main 2020 - 3 September (Morning)

The Kjeldahl method of Nitrogen estimation fails for which of the following reaction products?



(A) (a), (c) and (d)

(B) (a) and (d)

(C) (c) and (d)

(D) (b) and (c)

Q5 JEE Main 2020 - 6 September (Morning)

In an estimation of bromine by Carius method, 1.6g of an organic compound gave 1.88g of AgBr. The mass percentage of bromine in the compound is (Atomic mass, $Ag = 108$, $Br = 80$ g mol⁻¹)

Q6 JEE Main 2020 - 6 September (Evening)

Match the following:

Test/Method	Reagent
(i) Lucas Test	(a) $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$ / aq. KOH
(ii) Dumas method	(b) HNO_3 / AgNO_3
(iii) Kjeldahl's method	(c) CuO/CO_2
(iv) Hinsberg test	(d) Conc. HCl and ZnCl_2
	(e) H_2SO_4

- (A) (i) – (b), (ii) – (d), (iii) – (e), (iv) – (a)
(B) (i) – (d), (ii) – (c), (iii) – (e), (iv) – (a)
(C) (i) – (b), (ii) – (a), (iii) – (c), (iv) – (d)
(D) (i) – (d), (ii) – (c), (iii) – (b), (iv) – (e)

Q7 JEE Main 2020 - 7 January (Evening)

A chromatography column, packed with silica gel as stationary phase, was used to separate a mixture of compounds consisting of (A) benzanilide (B) aniline and (C) acetophenone. When the column is eluted with a mixture of solvents, hexane : ethyl acetate (20 : 80), the sequence of obtained compounds is :

- (A) (B), (C) and (A)
(B) (A), (B) and (C)
(C) (B), (A) and (C)
(D) (C), (A) and (B)

Answer Key

Q1 (A)

Q2 (A)

Q3 (C)

Q4 (C)

Q5 (50)

Q6 (B)

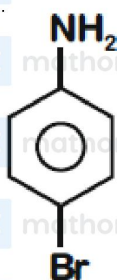
Q7 (D)

Q1

Kjeldahl method is not applicable to nitro or diazo groups present in the ring, as nitrogen atom can't be converted to ammonium sulfate under the reaction conditions.

Q2

Given 0.172g organic compound showed presence of 0.08g Bromine or 172g organic compound will show 80g Bromine which matches, exactly with option (1)



Contain 80g Bromine
M.W. = 172

Q3

Distillation under reduced pressure is done for separating glycerol in soap industries.

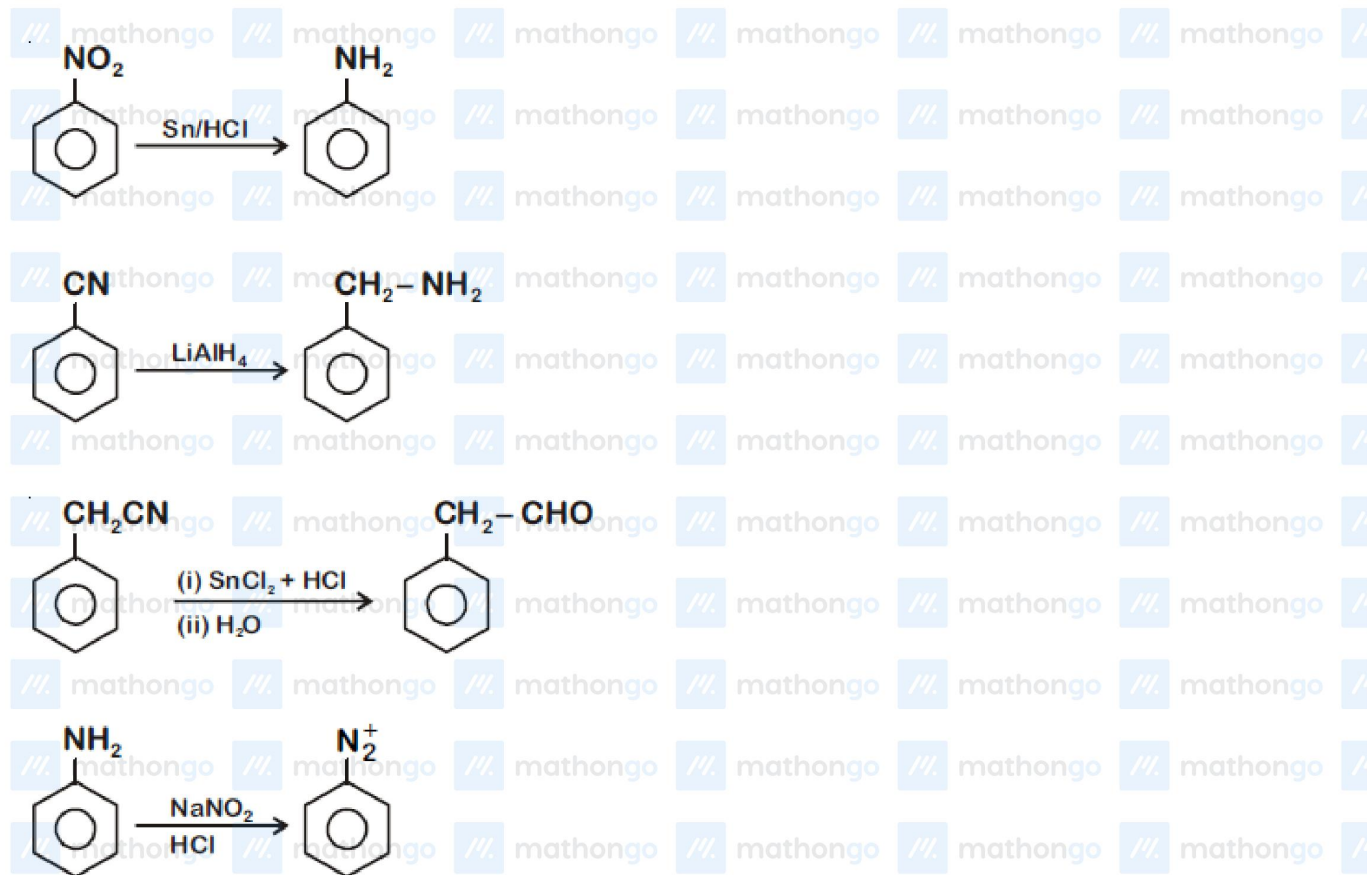
Q4

Purification Of Organic Compounds

Hints & Solutions

JEE Main 2020 Chapterwise

Chemistry



Nitro and diazo compounds do not give Kjeldahl estimation of Nitrogen.

Q5

Mass of organic compound = 1.6gm

Mass of AgBr = 1.88gm

Moles of Br = Moles of AgBr = $\frac{1.88}{188} = 0.01$

Mass of Br = $0.01 \times 80 = 0.80\text{gm}$

% of Br = $\frac{0.80 \times 100}{1.60} = 50\%$

Q6

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Hints & Solutions

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Chemistry

Lucas reagent — Conc. HCl/ZnCl_2

Dumas method — CuO/CO_2

Kjeldahl's method — H_2SO_4

Hinsberg test — $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}/\text{aq. KOH}$

Q7

More polar compound will come out first

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